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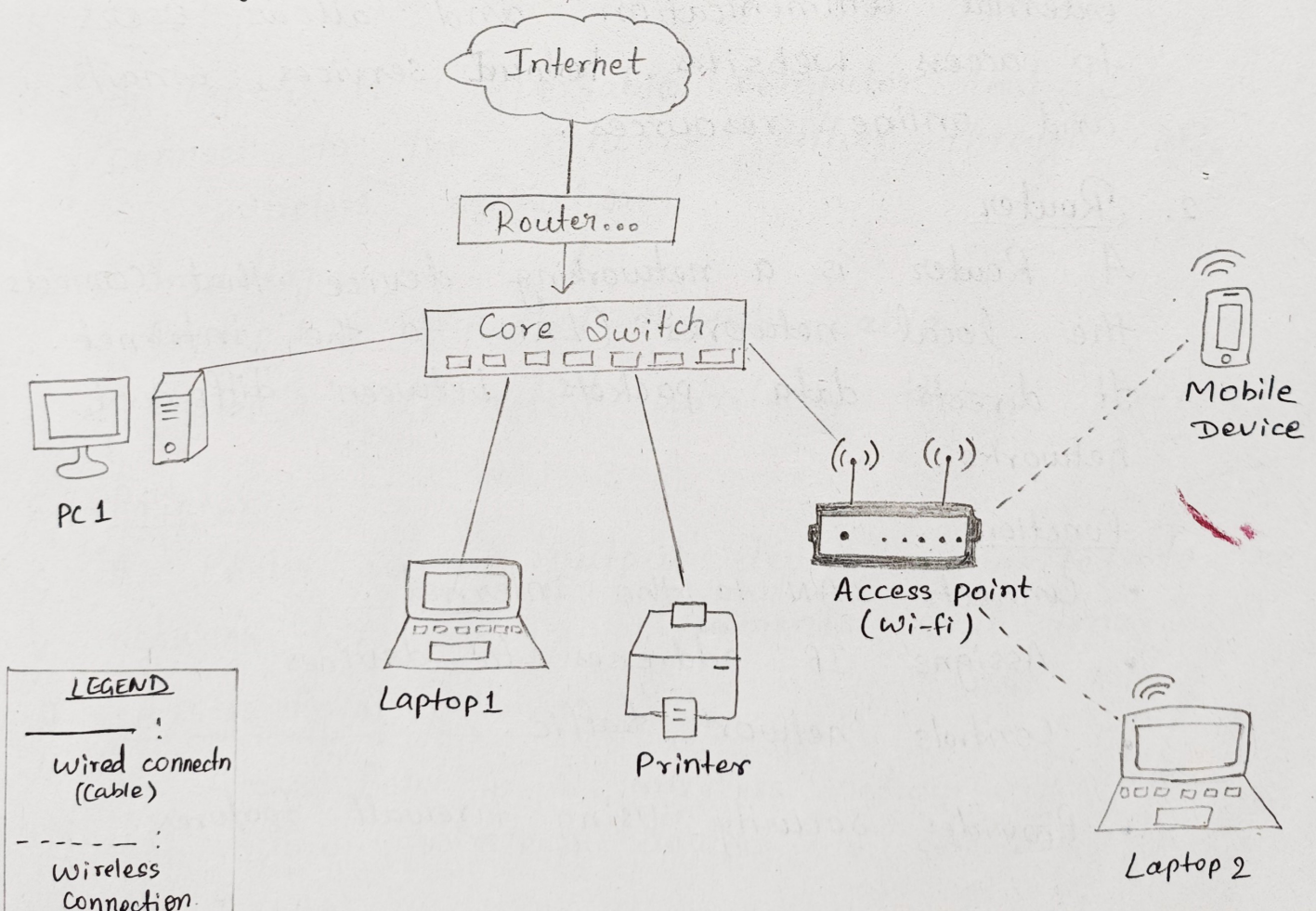
1. Draw the Network Topology :
   - Identify and draw the LAN topology of your organization (e.g. Star, Bus, Mesh).
   - Clearly label all components such as Router, Switch, Access point, and end devices (PCs, laptops, etc.).

→ Network Topology (Star Topology):

Aim : To study and draw the LAN Network topology used in our organization.

↝ The standard Local Area Network (LAN) architecture uses a Hierarchical Star Topology.

↝ Diagram of ☆ STAR TOPOLOGY ☆



→ Introduction :-

A star topology is a network design in which all devices are connected to a central device such as a switch or hub. It is one of the most commonly used topologies in modern Local Area Networks (LANs) because it is reliable, easy to manage, and simple to expand.

→ Components used in the diagram :

1. Internet

The Internet is a global network that connects millions of computers and devices worldwide. In the diagram, the internet provides external communication and allows users to access websites, cloud services, emails, and online resources.

2. Router

A Router is a networking device that connects the local networks (LAN) to the internet. It directs data packets between different networks.

Functions

- Connects LAN to the internet.
- Assigns IP addresses to devices.
- Controls network traffic.
- Provides security using firewall features.

3. Core Switch

A core switch is the central device in a star topology. All wired devices are connected to it. It receives and forwards data to the correct destination device.

Importance:

If the core switch fails, communication between connected devices stops.

4. PC (Personal Computer)

A PC is an end-user device used for various tasks such as browsing, document creation, programming, and communication.

5. Laptop

A Laptop is a portable computer that can connect to the network either through cable or wireless connection.

6. In diagram :-

- Laptop 1 : Connected through cable.
- Laptop 2 : Connected wirelessly through Wi-Fi.

6. Printer

A Printer is an output device connected to the network for printing documents and images.

7. Access Point (Wi-Fi ⌔):

An Access Point is a wireless networking device that allows wireless devices to connect to the LAN.

8. Mobile Device

A Mobile Device refers to smartphones or tablets connected wirelessly to the network.

Connection Type:

Wireless connection through Access point.

→ Advantages of star Topology :-

1) Easy to install and manage.
2) Failure of one device does not affect others.
3) Easy to trobleshoot.
4) Better network performance.
5) Easy to add new devices.

→ Conclusion :

The given diagram represents a Star Topology, LAN Network where all devices communicate through a central Core Switch. Wired and wireless devices can easily share resources and access the internet efficiently. This topology is widely used in schools, offices, and organizations because of its reliability and ease of maintenance.

2. Identification of Network Devices in LAN :

(a) Introduction :

A Local Area Network (LAN) is a network that connects computers and other devices within a limited area such as a home, school, office or laboratory. Different networking devices are used in a LAN to enable communication, data sharing, internet access, and network management. Each device performs a specific function to ensure smooth operation of the network.

(b) List of Networking Devices in LAN :

1. Network Interface Card (NIC)
2. Hub
3. Switch
4. Router
5. Bridge
6. Repeater
7. Modem
8. Wireless Access Point (WAP)
9. Gateway
10. Firewall

(c) Functions of Each Device :

1. Network Interface Card (NIC):

Function

- It Connects a computer to the network.
- Provides a unique MAC address for communication.

- Sends and receives data over the LAN.

2) Hub

- Connects multiple devices in a network.
- Broadcasts data to all connected devices.
- Helps in basic network communication.

3) Switch

- Connects device within a LAN.
- Sends data only to the target device.
- Reduces network traffic and improves speed.

4) Router

- Connects different networks together.
- Routes data packets between networks.
- Provides internet access to LAN devices.

5) Bridge

- Connects two LAN segments.
- Filters unnessessary network traffic.
- Improves network performance.

6) Repeater

- Regenerates weak signals.
- Extends the distance of a network.
- Prevents signal loss during transmission.

7) Modem

- Converts digital signals to analog signals.
- Converts analog signals back to digital signal.
- Enables internet connectivity.

8) Wireless Access Point (WAP):
- Provides Wi-Fi connectivity.
- Connects wireless devices to the LAN.
- Expands wireless network coverage.

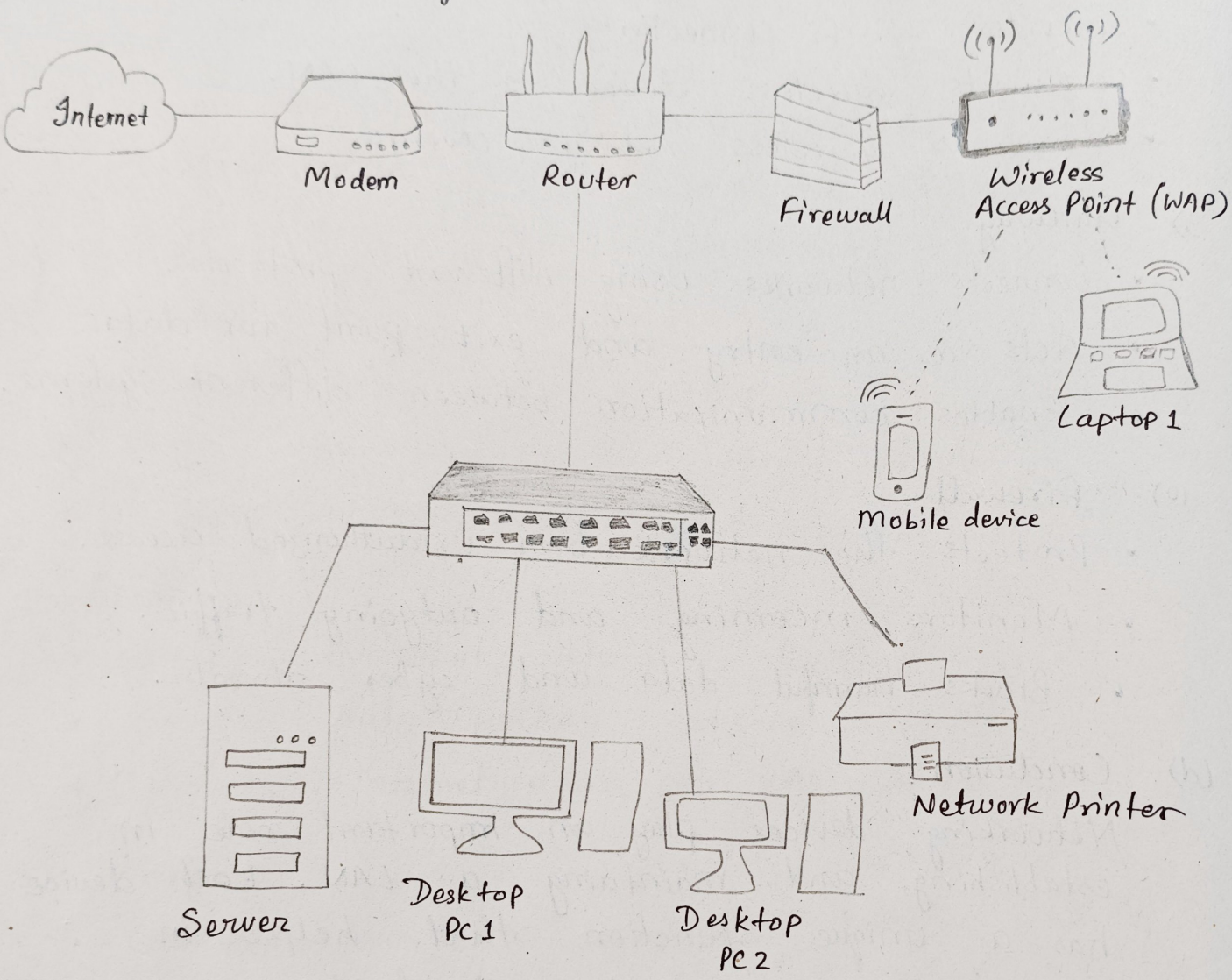
9) Gateway
- Connects networks using different protocols.
- Acts as an entry and exit point for data.
- Enables communication between different systems.

10) Firewall
- Protects the network from unauthorized access.
- Monitors incoming and outgoing traffic.
- Blocks harmful data and cyber threats.

(d) Conclusion:

Networking devices play an important role in establishing and maintaing a LAN. Each device has a unique function that helps in communication, security, signal transmission, and efficient data management within the network.

LAN Network Diagram :-



3. Analyze Connection Types:

In a Local Area Network (LAN), devices communicate using two major connection types: Wired (Ethernet) and Wireless (Wi-fi). Wired connections use Ethernet cables to transfer data between devices and networking equipment such as switches and routers. Wireless connections use radio waves through a Wi-fi router or access point, allowing devices to connect without cables.

In this experiment, the available network connection on the laptop was Wireless (Wi-fi) because the laptop does not have a built-in Ethernet ~~port~~ (RJ45) port. Therefore, practical testing of wired connection was not possible. Only wireless (Wi-fi) network analysis was performed practically.

A. Wireless (Wi-fi) Connection :-

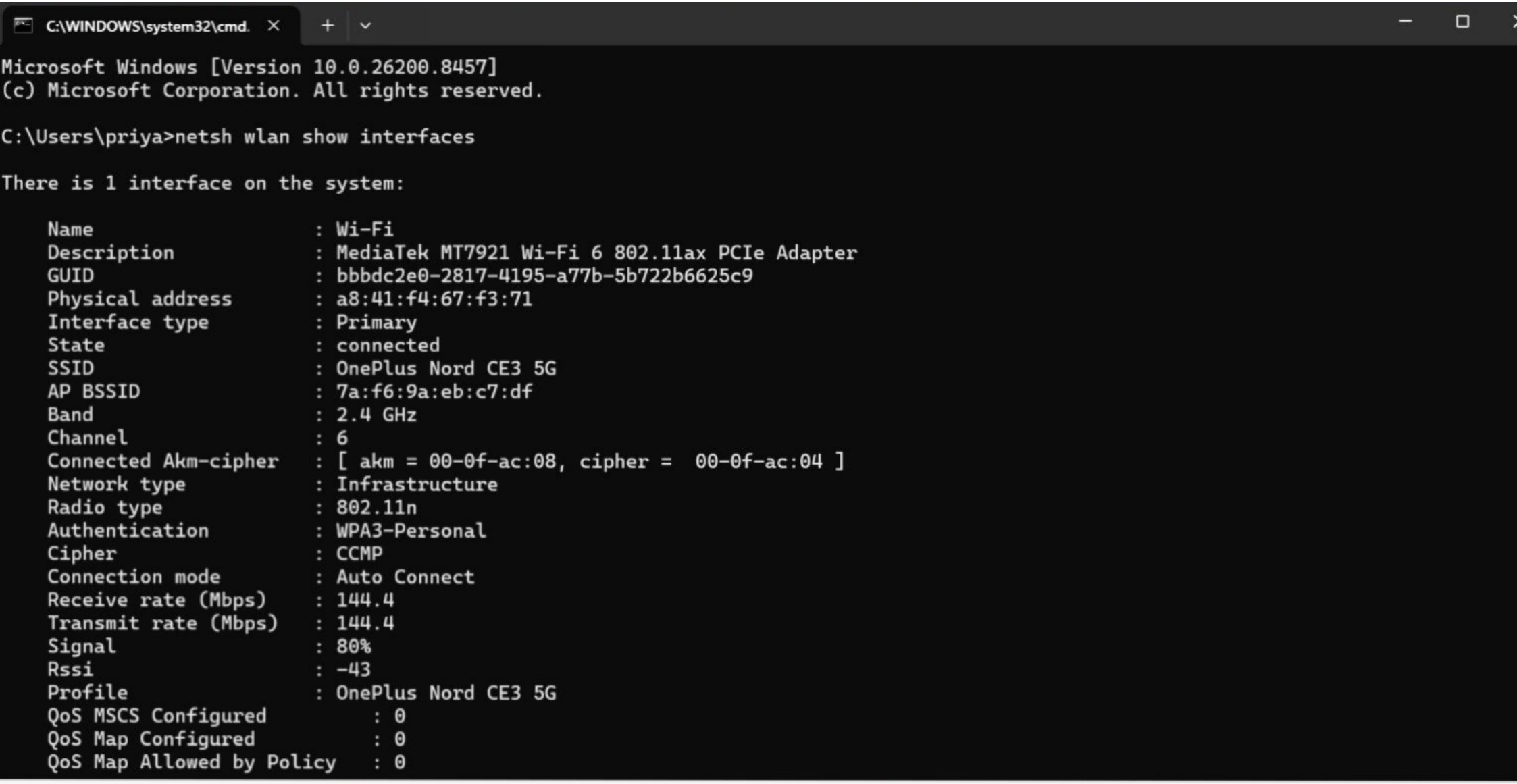
Step 1: Open Command Prompt
- a) Press Windows + R
- b) Type cmd
- c) Press Enter

Step 2: Check Wireless Network Details
- a) Type the following command:
  ~~nt~~ netsh wlan show interfaces
- b) Press Enter

Step 3: Observe the Output:
The command displays :-
- Wi-fi name (SSID) : Oneplus Nord CE3 5G

- Signal Strength : 80%
- Connection Status : connected
- Receive and transmit speed : 144.4 (Mbps)

→ Screenshot of Wireless Network Output :-


C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\priya>netsh wlan show interfaces
There is 1 interface on the system:
Name : Wi-Fi
Description : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
GUID : bbbdc2e0-2817-4195-a77b-5b722b6625c9
Physical address : a8:41:f4:67:f3:71
Interface type : Primary
State : connected
SSID : OnePlus Nord CE3 5G
AP BSSID : 7a:f6:9a:eb:c7:df
Band : 2.4 GHz
Channel : 6
Connected Akm-cipher : [akm = 00-0f-ac:08, cipher = 00-0f-ac:04]
Network type : Infrastructure
Radio type : 802.11n
Authentication : WPA3-Personal
Cipher : CCMP
Connection mode : Auto Connect
Receive rate (Mbps) : 144.4
Transmit rate (Mbps) : 144.4
Signal : 80%
Rssi : -43
Profile : OnePlus Nord CE3 5G
QoS MSCS Configured : 0
QoS Map Configured : 0
QoS Map Allowed by Policy : 0

Explanation

The laptop is connected to the network through Wi-Fi. Wireless networking provides mobility and ease of access because devices can connect without physical cables. It is commonly used for:

- Laptops
- Smartphones
- Tablets

However, wireless connections may experience:

- Signal interference
- Lower speed compared to Ethernet.
- Higher latency.

B. Wired (Ethernet) Connection:-

→ A wired connection uses an Ethernet cable to connect devices directly to networking equipment such as a switch or router. Ethernet provides:

- Stable communication
- Faster speed
- Lower latency
- Better security

In this system, Ethernet is currently not connected because the laptop does not have an active Ethernet (RJ45) port. Therefore, practical output could not be captured.

→ Common uses of Wired connection:

Ethernet connections are generally used for:

- Desktop computers
- Office networks

- Servers
- ~~Onling~~ Online gaming
- High speed data transfer.

→ Typical Ethernet Command

If Ethernet were available, the following command could be used:

```
ipconfig
```

or

```
Get-NetAdapter
```

These commands display:

- Ethernet adapter information.
- IP address
- Link status
- Network speed

Conclusion :-

The network analysis showed that the laptop currently uses a Wireless (Wi-Fi) connection for Internet access. Wireless networking provides flexibility and convenience for portable devices. Although Ethernet was not available on the system, wired connections generally offer better speed, reliability, and security compared to wireless networks.

4. Measure Network Speed :-

- Find the link speed of :
  - Wired Network (if available)
  - Wireless network

Objective : To measure the link speed of the Wireless (Wi-Fi) network using command prompt.

Introduction

Network speed refers to the rate at which data is transferred between devices in a network. The link speed can be measured for both Wired (Ethernet) and Wireless (Wi-Fi) connections using different Windows tools and commands.

In this experiment, the laptop was connected through Wireless (Wi-Fi) only. Ethernet connection was not avaiable during the experiment.

A. Wireless Network Speed (Wi-Fi) :

Step 1 : Open Command Prompt

1. Press Windows + R
2. Type cmd
3. Press Enter

Step 2 : Run the command

netsh wlan show interfaces

Press Enter

Step 3 : Observe the Output

The command displays :

- Signal strength
- Receive speed

- Transmit speed
- Wi-Fi adapter details

~~Output (Result)~~

Example (Output)

Receive rate (Mbps) : ~~866.7~~ 144.4
Transmit rate (Mbps) : 144.4
Signal : 80%.

Step4: Result

- Wireless Network Link Speed : 150 Mbps (example)
- The Wi-Fi connection was stable with good signal strength.

Step5: The tool / command Used:

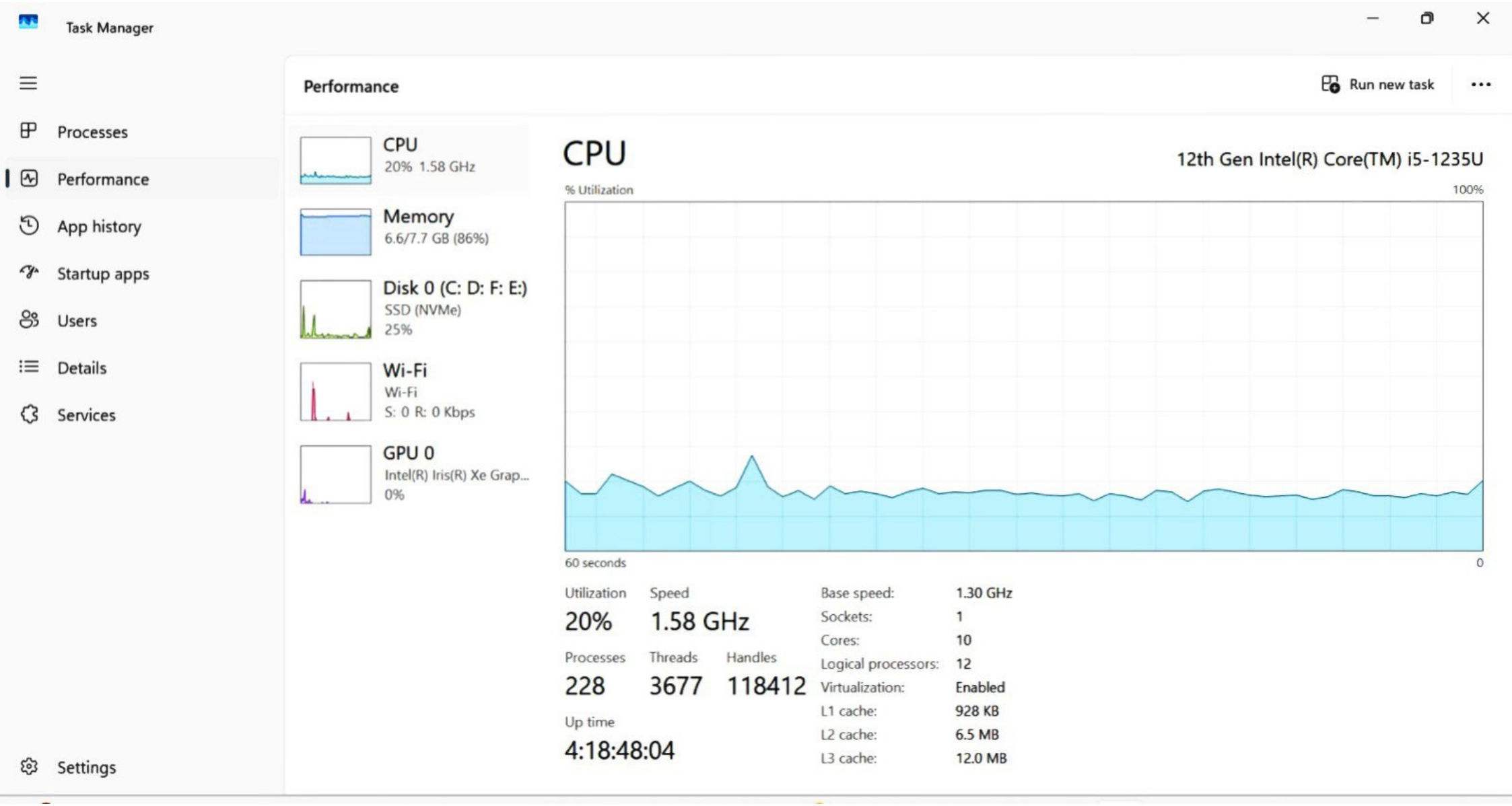
The following tools / commands were used:

- Command Prompt
- netsh wlan show interfaces
- Test Manager → Performance → Wi-Fi

Step6: Reference photo of Wi-Fi speed / output is given ~~above~~ below:-

Conclusion:

The network speed was measured using Windows networking tools and commands. The laptop was connected through a wireless network, and the Wi-Fi link speed was successfully observed using the netsh wlan show interfaces command. Ethernet testing could not be performed because no Ethernet (RJ45) port is available.


Task Manager
Processes
Performance
App history
Startup apps
Users
Details
Services
Settings
Performance
Run new task
CPU
20% 1.58 GHz
Memory
6.6/7.7 GB (86%)
Disk 0 (C: D: F: E:)
SSD (NVMe)
25%
Wi-Fi
Wi-Fi
S: 0 R: 0 Kbps
GPU 0
Intel(R) Iris(R) Xe Grap...
0%
CPU
12th Gen Intel(R) Core(TM) i5-1235U
% Utilization
100%
60 seconds
0
Utilization
20%
Speed
1.58 GHz
Processes
228
Threads
3677
Handles
118412
Up time
4:18:48:04
Base speed: 1.30 GHz
Sockets: 1
Cores: 10
Logical processors: 12
Virtualization: Enabled
L1 cache: 928 KB
L2 cache: 6.5 MB
L3 cache: 12.0 MB

5. Find System Network Details:

- Determine and record the foll. of your computer:
  - MAC Address
  - IP address (IPv4)
  - Subnet Mask
- Mention the command(s) used (e.g, ipconfig, getmac).

Introduction

Every computer connected to a network has some unique identification details like MAC Address, IP Address (IPv4), and Subnet Mask. These details help in identifying the device on a network and allow communication between different devices.

To find these details in a Windows computer, we use built-in commands like ipconfig and getmac.

Step-by-Step procedure

Step 1: Open Command Prompt

Step 2: Find IP Address and Subnet Mask

In command prompt, type:

ipconfig

- Press Enter
- Look For
  - IPv4 Add.
  - Subnet Mask

Example: In my computer, IPv4 Add. : 10.85.17.159
(Result) Subnet Mask : 255.255.248.0

Step3: Find MAC Address

- In Command prompt :, type:

  getmac

- Press Enter
- It will show the Physical Add. (MAC) Add.

Example: getmac output showing MAC address.

Network Details

- IP Address (IPv4) : 10.85.17.159
- Subnet Mask : 255.255.248.0
- MAC Address : A8-41-F4-67-F3-71

Commands Used

ipconfig → Used to display IP Add. and Subnet Mask.

getmac → Used to display MAC Address.

Conclusion :-

Using simple command-line tools, we can easily find important network details of a computer such as IP Add., subnet Mask, and MAC address. These details are essential for identifying and managing devices in a network.

6. Explain Observations:

- Breifly explain:
  - Diff. blw wired and wireless speed
  - Role of subnet Mask in networking.

1) Diff. blw wired and wireless speed :-

- Wired and Wireless connections differ mainly in speed, stability, and performance.

Wired (Ethernet)	Wireless (Wi-fi)
1. Provides higher and more stable speed.	1. Speed is slightly lower and may fluctuate.
2. Direct connection using cable.	2. Uses radio signals (no cable)
3. Less interference from other devices.	3. Can be affected by walls, distance, and devices.
4. Lower latency (better for gaming / streaming).	4. Higher latency compared to wired.
5. More reliable for heavy usage.	5. More convenient but less consistent.

2. Role of Subnet Mask in Networking :-

- A subnet Mask is used to divide an IP address into :
  - Network part
  - Host part
- It helps in identifying :
  - which network the device belongs to
  - which devices are in the same network.
- It allows communcation between devices by :
  - seperating local network traffic from external traffic.
- Example :
  - IP Address : 192.168.1.10
  - Subnet Mask : 255.255.255.0
    → Shows that device is in the same local network (192.168.1.x).

3. Conclusion:-

Wired connections are faster and more stable compared to wireless connections, while Subnet Mask plays an important role in organizing and managing network communication.